DEEP DOUBLE DESCENT: A REDUCED REPRODUCTION

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Assignment 2: Replication Track
https://github.com/HP2706/ATDL-Group-Work

ABSTRACT

We reproduce selected experiments from "Deep Double Descent: Where Bigger Models and More Data Hurt" by Nakkiran et al. We reuse the released vision model definitions, implement training and evaluation, and compare our CIFAR-10/100 results with the authors' released measurements. Vision experiments use one seed per setting and reduced training horizons: 400 Adam epochs or 50,000 SGD updates, corresponding to one tenth of the stated horizons, or one twentieth for the CIFAR-100 CNN. Noisy width sweeps qualitatively reproduce model-wise double descent and shifts of the interpolation threshold with label noise, augmentation and sample size. Small, intermediate and large ResNet18 models exhibit decreasing, U-shaped and double-descent test-error trajectories, respectively, although the large model's final error remains above its first minimum. The CNN plateau is not reproduced at the authors' widths; unequal training progress under the reduced update budget is a plausible explanation. Clean CIFAR-100 peaks remain unconfirmed. At the original training horizon, IWSLT'14 experiments show non-monotonic test loss with width and sample size, but do not reproduce the width-sweep region where 18,000 examples perform worse than 4,000. Missing implementation details limit exact reproduction of the translation results. These results provide meaningful qualitative but incomplete support for the selected findings.

1. INTRODUCTION

Classical learning theory predicts a U-shaped bias-variance trade-off, where larger models overfit beyond some complexity. Modern networks contradict this, generalising better as they grow far beyond the number of training samples. Which regime applies matters in practice. It affects whether to enlarge a model, train it longer, stop early or collect more data.

Nakkiran et al. [1] reconcile the two views through the *effective model complexity* (EMC) of a training procedure: the largest number of samples it can fit to near-zero training error (Section 2). They hypothesize that test error behaves atypically when the EMC is close to the number of training samples n. Varying the width, the training time or n moves a model through this critical regime. The results are *model-wise* and *epoch-wise* double descent, and *sample-wise* non-monotonicity, where more data can hurt. The authors demonstrate these effects for CNNs, ResNets and Transformers on CIFAR-10/100 and machine translation. We organize our reproduction around what we believe are the five main empirical findings described in the paper, labelled C1–C5 below:

- C1 Model-wise double descent exhibits a test-error peak near the interpolation threshold [1, Section 2, p. 6; Section 5].
- C2 Adding label noise or data augmentation, or increasing the training-set size, shifts the interpolation threshold and the associated test-error peak towards larger models [1, Section 5, p. 8].
- C3 As training proceeds, small, intermediate and large models can exhibit monotonically decreasing, U-shaped and double-descent test-error trajectories, respectively [1, Section 6, p. 10].
- C4 Near the critical regime, increasing the training-set size may provide little benefit and, in some settings, worsen test performance [1, Section 7, pp. 11–12].
- C5 Optimal early stopping suppresses double descent in many tested settings, but exceptions remain [1, Section 8, p. 15].

We assess C1, C3 and C4 as central findings, with C2 and C5 providing supporting evidence.
We implement training and evaluation for vision models and neural translation models, comparing our results with the authors' measurements. Appendix E presents paired plots for each reproduced figure.

2. BACKGROUND

A training procedure $\mathcal{T}$ maps a training set $S = \{(x_i, y_i)\}_{i=1}^n$ to a classifier $\mathcal{T}(S)$. For a data distribution $\mathcal{D}$ and error toler-

ance $\epsilon > 0$, the EMC is

$$\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) = \max\{n \mid \mathbb{E}_{S\sim\mathcal{D}^n}[\mathrm{Error}_S(\mathcal{T}(S))] \leq \epsilon\}, \quad (1)$$

Here, Error_S denotes training error. Following the authors' heuristic choice, we use $\epsilon = 0.1$ to identify the empirical training-error threshold. They distinguish three regimes. If $\mathrm{EMC} \ll n$ (under-parameterized) or $\mathrm{EMC} \gg n$ (over-parameterized), any change that increases the EMC decreases test error. If $\mathrm{EMC} \approx n$ (critically parameterized), such a change may either increase or decrease it. The *interpolation threshold* is the point where $\mathrm{EMC} = n$. Since the EMC cannot be computed directly, we follow the authors and estimate the threshold as the smallest width $\hat{k}_{\mathrm{interp}}$ whose final training error on the (possibly noisy) training labels is at most ϵ.

The EMC framework connects changes in width, training time and sample size to C1, C3 and C4. Greater width or longer training can increase EMC, while larger datasets, noise and augmentation can shift the interpolation threshold towards larger models (C2).

3. METHODOLOGY AND EXPERIMENTAL DESIGN

3.1. Vision setup

The authors' public training histories show that several target effects emerge well before the full training horizon. We therefore shortened training to cover more experimental conditions within the same compute budget. This choice targets qualitative reproduction without assuming convergence; comparisons at intermediate training stages remain subject to uncertainty in the authors' logging intervals.

We reuse the released width-scaled ResNet18 and five-layer CNN definitions and implement training and evaluation in PyTorch. CIFAR-10/100 sweeps use one seed, sparse width grids and selected label-noise rates of 0, 10 and 20%. Adam runs last 400 epochs and SGD runs 50,000 updates, one tenth of the stated horizons except for the CIFAR-100 CNN, which uses one twentieth.

We evaluate clean test error and training error against fixed, possibly corrupted labels. Preprocessing, optimizer settings and differences from the original protocol, including width grids and sample sizes, are specified in Table 4. Appendix B maps the experiments to the reproduced figures.

3.2. Translation setup

We reproduce IWSLT'14 German–English experiments using a six-layer encoder–decoder Transformer. The width sweep uses 25 embedding dimensions from 8 to 200 at 4,000 and 18,000 training pairs; the sample-size sweep uses widths 64 and 80 at 2,000 to 50,000 pairs. Both retain the stated 80,000-update horizon. We use Adam with warmup and inverse-square-root decay, BPE preprocessing and length-based batches of up to 4,096 tokens. Table 5 records the full configuration and unresolved details of the authors' setup. We compare the resulting test-loss curves in Figures 5 and 13.

4. RESULTS

4.1. Model-wise double descent

Our ResNet18 sweeps reproduce a local test-error peak followed by a second descent under 10% and 20% label noise on both CIFAR-10 and CIFAR-100 (Figure 6). The noise-free curves decrease with width; increasing label noise shifts the training-error threshold and generally the test-error peak towards larger widths (Figure 6). These results qualitatively support C1 and the label-noise component of C2, although our sparse width grid limits how precisely the peak can be located relative to the threshold.

Without augmentation, the CIFAR-10 curves on noisy data show a pronounced peak followed by decreasing test error (Figure 7). With augmentation, fitting the training labels requires larger widths, and the test-error's rise occurs further to the right (Figure 7). However, the augmented 20%-noise curve shows only a slight decline after its peak within the displayed range: test error rises from 21.74% at width 16 to 28.11% at width 48, then falls slightly by width 64 after 50,000 updates (Figure 7). This supports a shift in the critical region under augmentation (C2), while providing weaker evidence for a completed second descent (C1; Figure 7). The noise-free CIFAR-100 CNN peak reported by the authors is not resolved in our reproduction (Figure 9). For clean CIFAR-10, test error decreases with width under both SGD and Adam (Figure 8). This qualitative agreement spans both training procedures, but their different budgets, 50,000 SGD updates versus 400 Adam epochs, prevent isolating the optimizer's effect.

The minimum observed test-error envelope suppresses the opening comparison's peak, consistent with C5 (Figure 1). This is an oracle assessment using test evaluations, not validation-based early stopping.

4.2. Epoch-wise double descent

We summarize each trajectory by its first minimum $E(t_1)$, subsequent peak $E(t_p)$ and final error $E(T)$. The weak form of the large-model claim requires $E(T) < E(t_p)$, i.e. a second descent; the strong form requires $E(T) < E(t_1)$. All three C3 regimes appear within our 400 epochs (Table 1; Figure 2). The $k = 3$ error decreases monotonically to 28.9%. The $k = 12$ error reaches 20.7% at epoch 80 and then rises to 29.7% at epoch 400, still increasing. The $k = 64$ error falls to 15.1% at epoch 20, peaks at 26.1% at epoch 70 and descends to 20.9% at epoch 400. Thus the weak form holds, but the final error remains 5.8 percentage points above the first minimum (Figure 2; Table 1).

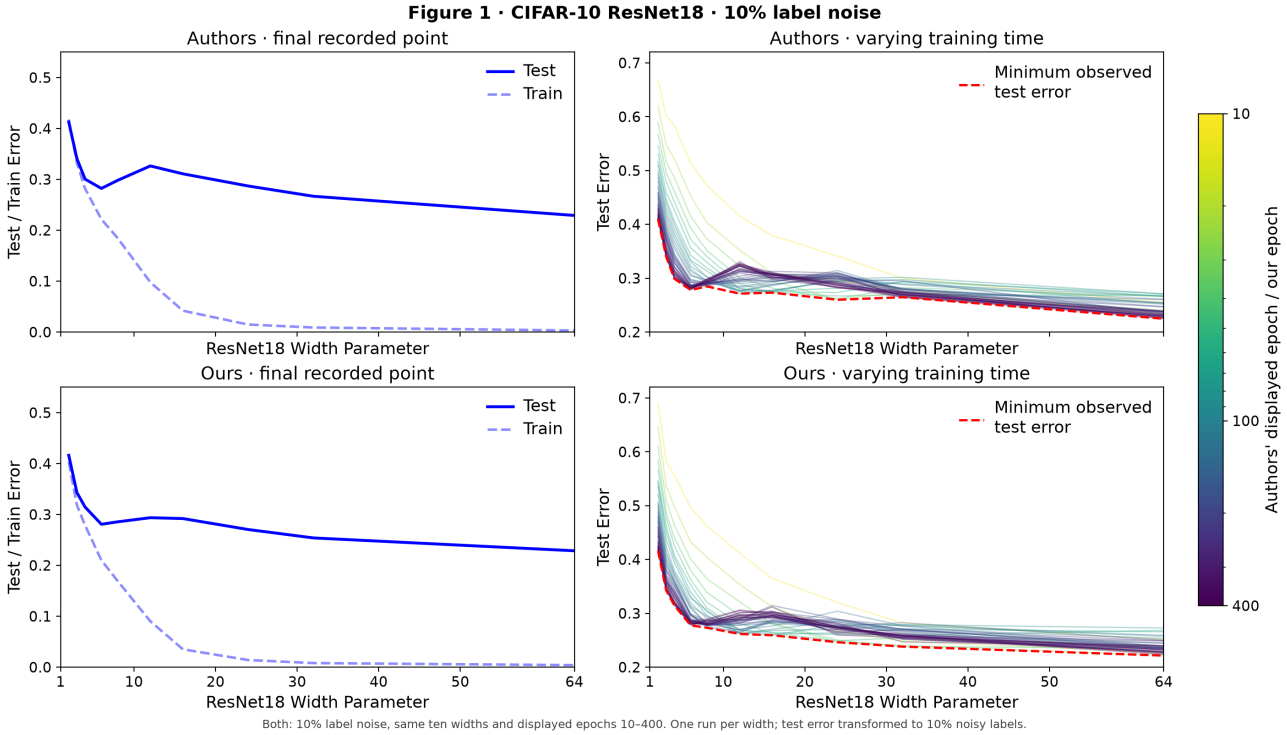


Fig. 1. Noise-matched adaptation of the opening figure. The original opening figure uses 15% noise and 4,000 epochs [1]. We use 10%-noise CIFAR-10 ResNet18 measurements above our 10%-noise results. Both panels use the same ten widths, 400 displayed epochs, and 10%-noise. Red curves show the minimum observed test error over the displayed evaluations.

k	Paper	$E(t_1)$	$E(t_p)$	$E(T)$	A: 400 / 4,000
3	monotone	–	–	28.9	29.6 / 24.6
12	U-curve	20.7 (80)	29.7 (400)	29.7	34.1 / 34.6
64	DD	15.1 (20)	26.1 (70)	20.9	21.9 / 19.6

Table 1. Epoch-wise test error (%), CIFAR-10 ResNet18, 20% label noise, one seed. $E(t_1)$: first minimum; $E(t_p)$: subsequent peak (epoch in parentheses); $E(T)$: ours at epoch 400. A: authors' released run at epochs 400 and 4,000.

The authors train for 4,000 epochs. Their released $k = 64$ trajectory is at 21.9% at epoch 400 and reaches 19.6% at epoch 4,000, so the second descent continues well beyond our horizon (Figure 2; Table 1). The stronger result of finishing below the first minimum is demonstrated at 10% noise in the paper's Figure 10 [1]; that condition is untested in our three-width ResNet comparison. Our width-128 CNN trajectories likewise show a second descent with 20% noise but not with clean labels (Figure 11).

4.3. Sample-wise non-monotonicity, CNN

Increasing the sample size shifts both the training-error threshold and the test-error peak towards larger CNN widths (C2; Figures 12 and 14). Table 2 estimates the 10% training-error crossing by linear interpolation between widths. Our thresholds increase from 17 to 27 to 56, compared with 15, 23 and 34 in the authors' grid; the peaks move in the same direction (Figure 12; Table 2).

The sample-wise experiment asks a different question from the moving peak: at a fixed width, does increasing the dataset improve test error? The authors' CNN curves contain a region where additional data provides little benefit (Figure 12). The CNN plateau (C4) is not reproduced at the authors' widths. Increasing sample size from 12,500 to 50,000 reduces their test error by 1.9 percentage points at width 32, compared with 10.1 points in our runs. At width 64, our reduction is 3.3 points (Figure 12; Table 2). The 12,500- and 25,000-sample grids have no widths between 32 and 64, limiting localization of any shifted plateau (Figures 12 and 14).

4.4. Translation results

The translation width sweeps exhibit peaks followed by decreasing loss, qualitatively supporting the width-dependent pattern in C1, but their peaks occur at smaller dimensions than the authors'. The 18,000-example curve remains below the 4,000-example curve throughout, so this experiment does not reproduce the reported region where more data hurts (Fig-

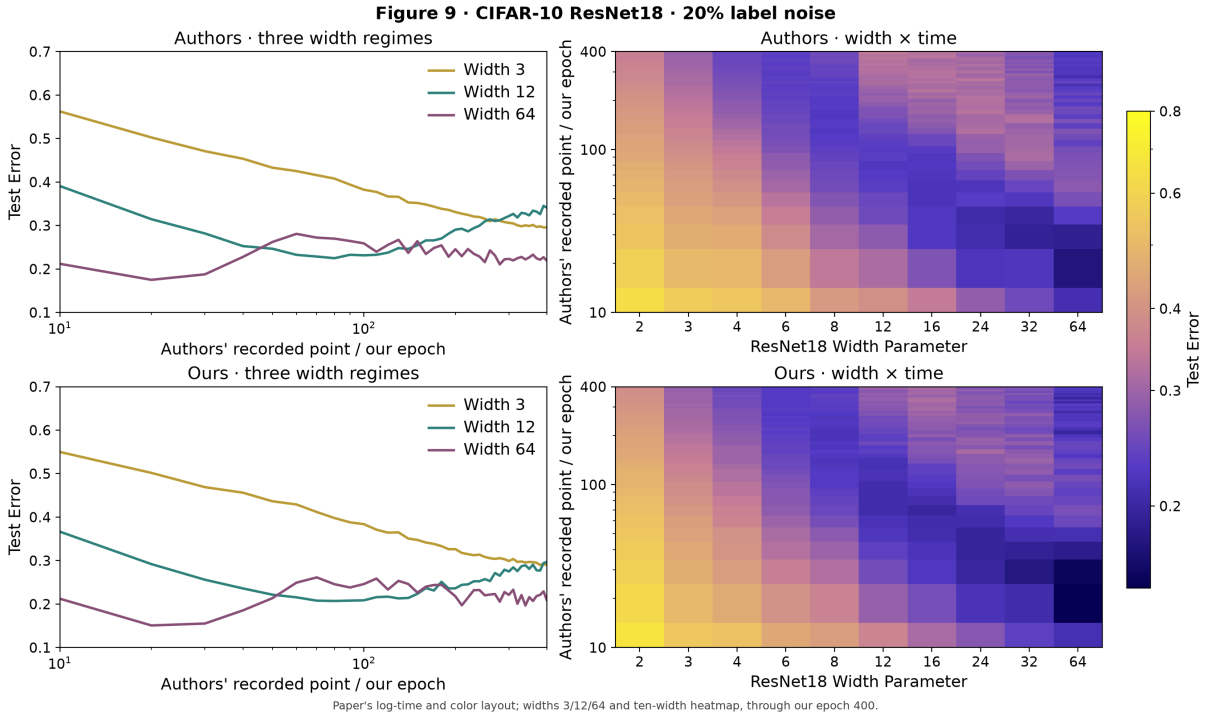


Fig. 2. Test error over training for ResNet18 widths $k = 3, 12, 64$ on CIFAR-10 with 20% label noise: authors' released runs (top, 4,000 epochs) and ours (bottom, 400 Adam epochs, one seed). Authors' epochs follow their notebook convention (measurement index plus one); their logging cadence is undocumented, so the time axes are aligned only approximately.

n	Threshold A / O	Peak A / O	Test, $k = 32$ A / O	Test, $k = 64$ A / O
12,500	15 / 17	15 / 16	33.9 / 35.7	29.7 / 31.2
25,000	23 / 27	20 / 32	34.1 / 34.1	27.0 / 29.0
50,000	34 / 56	31 / 48	32.0 / 25.6	27.0 / 27.9

Table 2. Sample-wise comparison, CIFAR-10 CNN with 20% label noise; authors (A) and ours (O). Threshold: width where train error reaches 0.1, linearly interpolated. Peak: test-error maximum after the first minimum. Test error in % at the common widths either side of the authors' 50,000-sample threshold. Authors on their released 42-width grid, ours on eleven widths $k \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 64\}$.

ure 5).

The sample-size sweep provides partial support for C4. At width 80, loss rises between 4,000 and 6,000 examples and falls below its 4,000-example value only at 12,000; width 64 shows a weaker increase from 2,000 to 4,000. Peak locations and absolute losses differ from the authors' (Figure 13).

5. DISCUSSION AND LIMITATIONS

5.1. Effect of the reduced protocol

Noisy ResNet18 sweeps agree qualitatively with the authors' displayed 400-epoch results, while augmented CNNs remain imperfectly fitted and show only partial second descent (Figures 6 and 7). Missing clean CIFAR-100 peaks remain inconclusive because training duration, sparse grids and implementation differences are not isolated (Figure 9).

At epoch 400, our errors for $k = 3, 64$ are within one percentage point of the authors' (Table 1). Their $k = 64$ run improves another 2.3 points by epoch 4,000 (Figure 2). Our observed second descent supports C3 without establishing improvement over the early minimum.

The horizon also affects how a width is classified, because EMC grows with training time. At 400 epochs our 20%-noise threshold is $\hat{k}_{\text{interp}} = 16$, so $k = 12$ has not yet crossed the threshold (Figure 6). The authors' $k = 12$ error changes little between epochs 400 and 4,000 (34.1 to 34.6%), providing no evidence of a delayed second descent within their longer horizon (Figure 2; Table 1).

The fixed update budget also gives larger datasets fewer passes: our 12,500-, 25,000- and 50,000-example CNN runs receive 512, 256 and 128 epochs. The larger set's threshold is displaced most relative to the authors' (Table 2), so

unequal progress towards fitting is a plausible explanation for the missing plateau (C4; Figures 12 and 14). The moving peak survives the reduction, but the plateau depends on where the sample-size curves meet (Figures 12 and 14). A targeted follow-up would extend training for the largest set, holding other settings fixed, and add widths around its shifted threshold.

5.2. Implementation

We believe our codebase follows the authors' implementation, however since the codebase does not include the training code, we have to rely on the information from the paper, which sometimes we found to be quite sparse or even contradictory. For example in the translation task, the exact preprocessed data, subsets, seeds and batching configuration could not be recovered, moreover the paper describes using Adam, while the released codebase mentions using SGD. Thus we cannot exclude the possibility that some of the discrepancies between results are due to different training and evaluation settings, we include a table with all key training and evaluation decisions we made and how it compares to what is in the paper (Table 4,Table 5).

5.3. Statistical uncertainty

For all experiments, we use a single seed for computational reasons. However the paper for many experiments relies on 5 independent seeds. Thus a single-seed experiments cannot establish the stability of peak heights or small performance differences, and sparse grids limit threshold localization (Figures 6 and 12). We could not identify the five trials underlying the published CNN sample-wise panel; our comparison instead uses the released single-trial grid (Figure 12).

6. CONCLUSION

The central findings receive qualitative but incomplete support: the width sweeps on noisy data exhibit double descent (C1; Figures 6 and 5), the three epoch-wise regimes appear (C3; Figure 2), and translation shows sample-wise nonmonotonicity (C4; Figure 13). The CNN plateau and clean CIFAR-100 peaks remain unresolved (Figures 12 and 9). Supporting findings include shifts of the critical region (C2; Figures 6, 7 and 14) and suppression of the opening peak by oracle early stopping (C5; Figure 1). The results support the EMC framework's qualitative predictions, subject to reduced horizons, single seeds and implementation uncertainty.

7. REFERENCES

[1] Preetum Nakkiran, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever, "Deep double descent: Where bigger models and more data hurt," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2021, no. 12, pp. 124003, 2021.

[2] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, vol. 30.

A. REPRODUCIBILITY AND CONTRIBUTIONS

The assignment code and result organization are available at `https://github.com/HP2706/ATDL-Group-Work`. Detailed experimental coverage and deviations are reported in Appendices B and C.

A.1. Individual contribution

Hans Peter Lyngsøe: Implemented the vision training code and experiment orchestration. Conducted the model-wise double-descent experiments and wrote subsection 3.1 and subsection 4.1. Contributed to the Introduction, Discussion and the Conclusion.

Mathias Elmegaard Nielsen: Epoch-wise double descent (paper Figures 9 and 10). Ran and analyzed the CIFAR-10 ResNet18 trajectories for widths 3, 12 and 64 at 20% label noise as well as the width-128 CIFAR-10 CNN trajectories at 0% and 20% label noise (paper figure 10c). Defined the first-minimum, peak and final-error criteria for the three regimes and compared them with the authors' released runs at epochs 400 and 4,000. Built Table 1 and Figure 2. Wrote the Introduction, Background and subsection 4.2, and the epoch-wise parts of the Discussion. Contributed to abstract and general discussion sections.

Julian Barragan: sample-wise non-monotonicity (paper Figures 11(a) and 12). Ran the CIFAR-10 CNN subset experiments at 12,500 and 25,000 samples; compared them with the authors' released sample grid; built the comparison figures and Table 2, locating the interpolation threshold and test-error peak for each sample size; identified fewer training epochs for larger datasets under the fixed update budget as a plausible explanation for the missing plateau; found that the released grid holds a single trial whereas the published panel averages five, and that Appendix B.4 and Section 4 of the paper disagree on the training horizon; wrote subsection 4.3, the sample-wise discussion and the revision of C2.

Harald Franck: Initial plan was to try to reproduce the plot of Figure 3 and 11(b) in the paper and investigate the reproducibility of those. Implemented and ran the translation experiments, investigated reproducibility issues, produced comparison plots, and wrote about this process. Also wrote the translation experiments results and parts of the discussion.

B. EXPERIMENTAL COVERAGE

The translation experiments use IWSLT'14 German–English: the model-size sweep uses 25 embedding widths $d_{\text{model}} \in \{8, 16, \ldots, 200\}$ at 4,000 and 18,000 training pairs; the sample-size sweep uses $d_{\text{model}} \in \{64, 80\}$ and $n \in \{2{,}000, 4{,}000, \ldots, 50{,}000\}$. Both use 80,000 updates, Adam with warmup and inverse-square-root decay, and a 4,096-token batch budget (Section 3.2). Table 3 links each reproduced experiment to its comparison figure in Appendix E and distinguishes our coverage from the full published protocol.

Table 3: Coverage of the main paper's figures. Figure references point to our appendix comparisons. The final column describes differences or omitted conditions, rather than additional planned experiments.

Paper figure	Plot and coverage	Differences from the full reproduction
1–2	Figures 3 and 4: 10%-noise ResNet18 width sweeps and heatmaps through 400 epochs.	Original opening figures use 15% noise and 4,000 epochs; our grid is smaller and uses one seed. Figure 1 averages five trials.
3	Figure 5: IWSLT'14 Transformer model-size sweep, 25 embedding widths from 8 to 200, with 4,000 and 18,000 pairs.	Same selected widths, sample sizes and 80,000-update horizon; exact preprocessing, subsets and training configuration could not be recovered.
4	Figure 6: CIFAR-10 and CIFAR-100 ResNet18 at 0/10/20% noise, ten widths, epoch 400.	Smaller width grid, no CIFAR-10 5/15% conditions, and 400 rather than 4,000 epochs.
5	Figure 7: augmented and unaugmented CIFAR-10 CNN at 0/10/20% noise, through width 64, after 50,000 updates.	Smaller grid and 50,000 rather than 500,000 updates. Authors' saved measurement indices do not establish an exact time alignment.
6	Figure 8: clean, unaugmented CIFAR-10 CNN with SGD and Adam, eleven widths.	One seed; 50,000 SGD updates or 400 Adam epochs rather than 500,000 updates or 4,000 epochs.
7	Figure 9: clean CIFAR-100 CNN at eleven shared widths.	One seed and 50,000 updates rather than five seeds, 64 widths and 1,000,000 updates.
8	Outside the reproduced subset: full-data IWSLT'14 and WMT'14 model-size sweeps.	Our translation experiments instead cover the smaller IWSLT'14 subsets in Figures 3 and 11(b).
9	Figure 10: three ResNet18 width trajectories and a ten-width heatmap through epoch 400.	Smaller width grid and 400 rather than 4,000 epochs.
10(c)	Figure 11: width-128 CIFAR-10 CNN trajectories at 0/20% noise, through 50,000 updates.	ResNet18 panels are outside our coverage. CNN horizon is reduced from 500,000 updates; authors' time coordinates are saved measurement indices.
11(a)	Figure 12: 20%-noise CIFAR-10 CNN at 12,500/25,000/50,000 examples and eleven common widths.	One seed and 50,000 updates; published panel averages five trials. Our additional 10%-noise subset runs lack a matching released source.
11(b)	Figure 13: IWSLT'14 sample-size sweep from 2,000 to 50,000 pairs at embedding widths 64 and 80.	80,000 updates; exact subsets, preprocessing and batching cannot be matched to the authors' configuration.
12	Figure 14: 20%-noise CNN heatmaps and slices at three sample sizes and eleven common widths.	Three rather than nine sample sizes, eleven rather than 42 widths, and 50,000 rather than 500,000 updates.

C. COMPARISON OF EXPERIMENTAL SETTINGS

Table 4 compares the completed vision experiments one setting at a time. Page numbers refer to the 2021 journal article (32 pages), not an earlier arXiv version [1]. Released code paths refer to commit `0d653296`. The release contains model definitions, plotting notebooks, and a `README.md` linking result archives; it does not provide the vision training loop or data loader. Where neither the paper nor those files specify a detail, we identify it as unstated rather than infer a setting from the results.

Table 4: Our model, data, and training settings, their source locations, and differences from the stated protocol.

Setting	Our setup	Paper setting and source location	Difference
Model			
Vision architectures	Released ResNet18 and five-layer CNN; widths $[k, 2k, 4k, 8k]$	Same architectures: p. 17, Appendix B.1. Code: `models/resnet18k.py` and `models/mcnn.py`.	Same definitions
Initialization	Random, default PyTorch initialization	Default PyTorch initialization: p. 17, Appendix B.2, "Initialization". Model files above instantiate PyTorch layers.	Same rule
Width grids	ResNet18: $k \in \{2, 3, 4, 6, 8, 12, 16, 24, 32, 64\}$. CNN: $k \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 64\}$, with augmented full-data SGD sweeps also including $36, 40, 48, 96, 128$	Width sweeps: p. 17, Appendix B.1; exact arrays in linked archives, e.g. `cifar10-resnet18k-50k-adam/ks` and `pct-cifar10-mcnn-50000-p0-sgd-big/ks`. `README.md` supplies archive links.	Fewer widths
Data			
Pixel preprocessing	Divide by 255; no channel normalization	Not stated in Section 4 (p. 8) or Appendix B.2 (pp. 17–18). No released data-loader file.	Not stated
Label corruption	Fixed Bernoulli mask; uniformly incorrect labels; fixed across augmentations	Once-sampled, uniformly incorrect labels: p. 8, Section 4, "Label noise". Fixed labels across augmentations: p. 17, Appendix B.2. No released implementation.	Same rule; seed not stated
Noise settings	0, 10, 20%; Figure 1 compares 10% on both sides	Figure 4, p. 7, and experiment list in `README.md`; original Figure 1, p. 4, uses 15%.	Same selected rates; original Figure 1 differs
Augmentation	Padding 4, random crop 32, horizontal flip	RandomCrop(32, padding=4), RandomHorizontalFlip: p. 17, Appendix B.2, "Data-augmentation". No released loader.	Same operations
Subsampling	Seeded permutation; 12,500/25,000/50,000 training images	Sample sizes: Figure 11(a), p. 13. Random subsampling: p. 19, Appendix B.4, "Sample-wise nonmonotonicity". No released subsampling code.	Same sizes; permutation and seed not stated
Test evaluation	Clean test split; Figure 1 also transforms error to the expected noisy-test distribution	Clean test error except Figure 1: p. 8, Section 4. Transformation implemented in `intro_resnet_plot.ipynb`, "Simple Plot" and `plot`.	Same error convention
Train evaluation	Fixed training labels, without augmentation	Exact train-evaluation transform not stated in Appendix B.2 (pp. 17–18). No released evaluation loop.	Transform not stated
Training			

Continued on the next page

Table 4 (continued)

Setting	Our setup	Paper setting and source location	Difference
Loss	Cross-entropy	Cross-entropy: p. 17, Appendix B.2, "Loss function". No released training loop.	Same loss
Batch size	128	128: p. 18, Appendix B.2, "Batch size". No released loader.	Same size
Adam	LR 10^{-4}; $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$	Same settings: p. 17, Appendix B.2, "Optimization: Adam". No released optimizer code.	Same settings
SGD	$\eta_t = 0.1/\sqrt{1 + \lfloor t/512 \rfloor}$; momentum 0	Same prescription: p. 18, Appendix B.2, "SGD" and "Learning rate schedule descriptions". No released optimizer code.	Same prescription
Regularization	No weight decay or dropout	None unless stated otherwise: p. 17, Appendix B.2, "Regularization".	Same setting
Training horizon	Adam 400 epochs; SGD 50,000 updates	Adam 4,000 epochs; SGD 500,000 steps: p. 8, Section 4. CIFAR-100 CNN uses 1 million steps: Figure 20, p. 25, Appendix E.2.1.	10% of stated horizon; 5% for CIFAR-100 CNN
Trials	One seed (0) per condition	Five trials for Figures 1, 7 and 11(a), bottom panel: p. 19, Appendix B.4. Figure 1 averaging is in `intro_resnet_plot.ipynb`.	One trial rather than five
Precision	FP32	Arithmetic precision not stated in Section 4 or Appendix B.2. No released training loop.	Not stated
Evaluation cadence	Adam: 10 epochs; SGD: 1,250 updates	`intro_ocean_dynamics.ipynb` labels ResNet indices plus one as epochs. CNN logging interval is not established by the paper or release.	No confirmed common cadence
Run infrastructure	Our training loop and checkpoints	Checkpointing, concurrency and GPU scheduling not stated in Appendix B.2 or provided in the released files.	Our implementation

Table 5: Translation configuration and limits on matching the authors' setup. Paper settings refer to Nakkiran et al. [1]; the optimizer prescription follows Vaswani et al. [2].

Setting	Our setup	Original specification or uncertainty
Architecture	Fairseq v0.9.0 Transformer; six encoder and six decoder layers, eight heads, feed-forward width $4d_{\text{model}}$, sinusoidal positions, post-normalization	Transformer family specified; exact Fairseq version and full configuration not recoverable.
Data	IWSLT'14 German–English, BPE preprocessing	Exact preprocessed files, subsets and seeds not recovered.
Width sweep	$d_{\text{model}} = 8, 16, \ldots, 200$; 4,000 and 18,000 pairs	Selected dimensions and sample sizes match the Figure 3 design.
Sample-size sweep	$d_{\text{model}} \in \{64, 80\}$; $n = 2{,}000, 4{,}000, \ldots, 50{,}000$	Reproduces the selected Figure 11(b) range.
Loss and regularization	10% label smoothing, no dropout	Same stated settings; paper plot labels alternate between perplexity and cross-entropy.
Optimizer	Adam, $(\beta_1, \beta_2) = (0.9, 0.98)$, $\epsilon = 10^{-9}$	Paper specifies Adam following Vaswani et al.; release describes SGD. We follow the paper.
Schedule	4,000 warmup steps; $\eta_t = d_{\text{model}}^{-1/2} \min(t^{-1/2}, t\, 4000^{-3/2})$	Warmup and inverse-square-root decay follow Vaswani et al.; exact author configuration not recovered.
Batching	Up to 4,096 tokens; length-based batches with variable sentence counts	Exact original batching strategy and token budget not recovered.
Horizon	80,000 gradient updates	Same stated horizon.

D. TEXT-LENGTH MEASUREMENT

We remove figures, tables, captions and references, then re-typeset the remaining content in the unchanged ISBI layout:

$$N(D) = \mathrm{pages}\big(L(P(D))\big).$$

We count main text separately from appendix prose; a partially filled final page counts as one page.

E. REPLICATION FIGURES: OURS COMPARED WITH THE AUTHORS

This section shows our replication results alongside measurements reconstructed from the authors' released data.

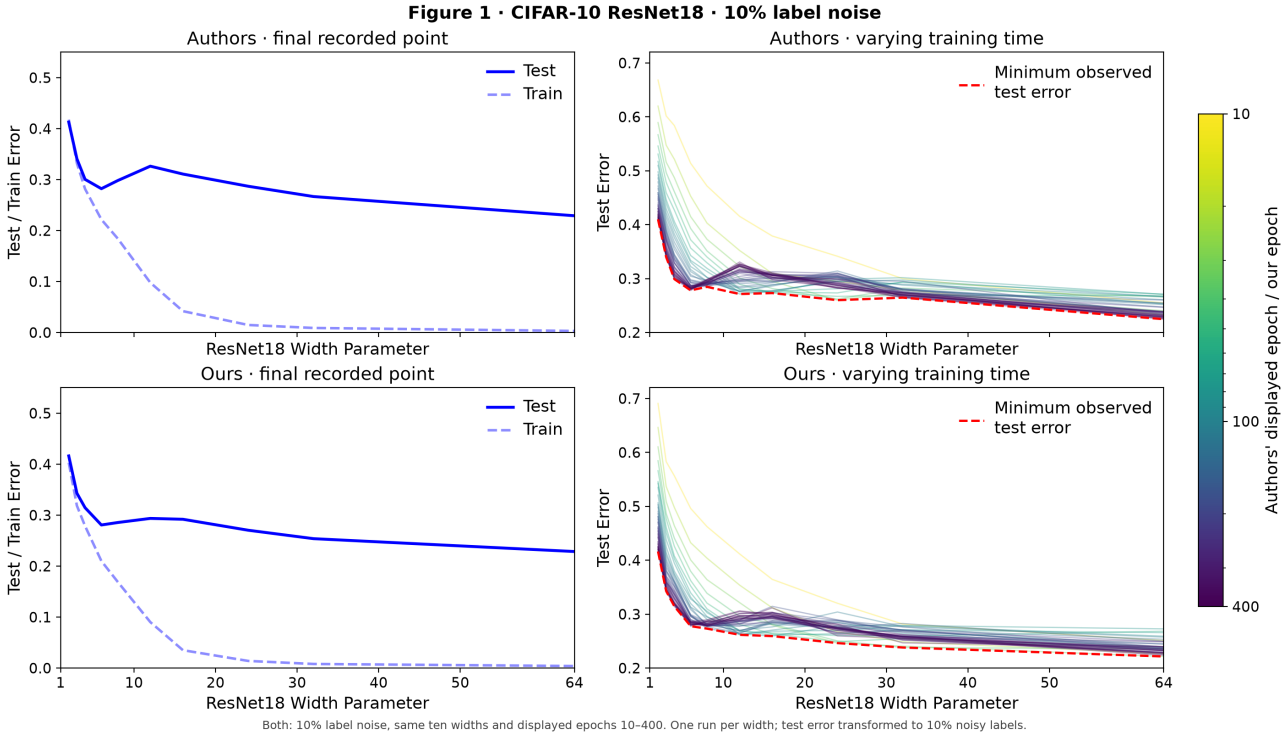


Fig. 3. CIFAR-10 ResNet18 width sweeps (paper Figure 1). Both authors and ours use 10%-noise CIFAR-10 ResNet18, the same ten widths and 400 displayed epochs, with one run per width. Authors' epochs follow the notebook convention; ours are recorded epochs. Test error uses the same expected noisy-test transformation.

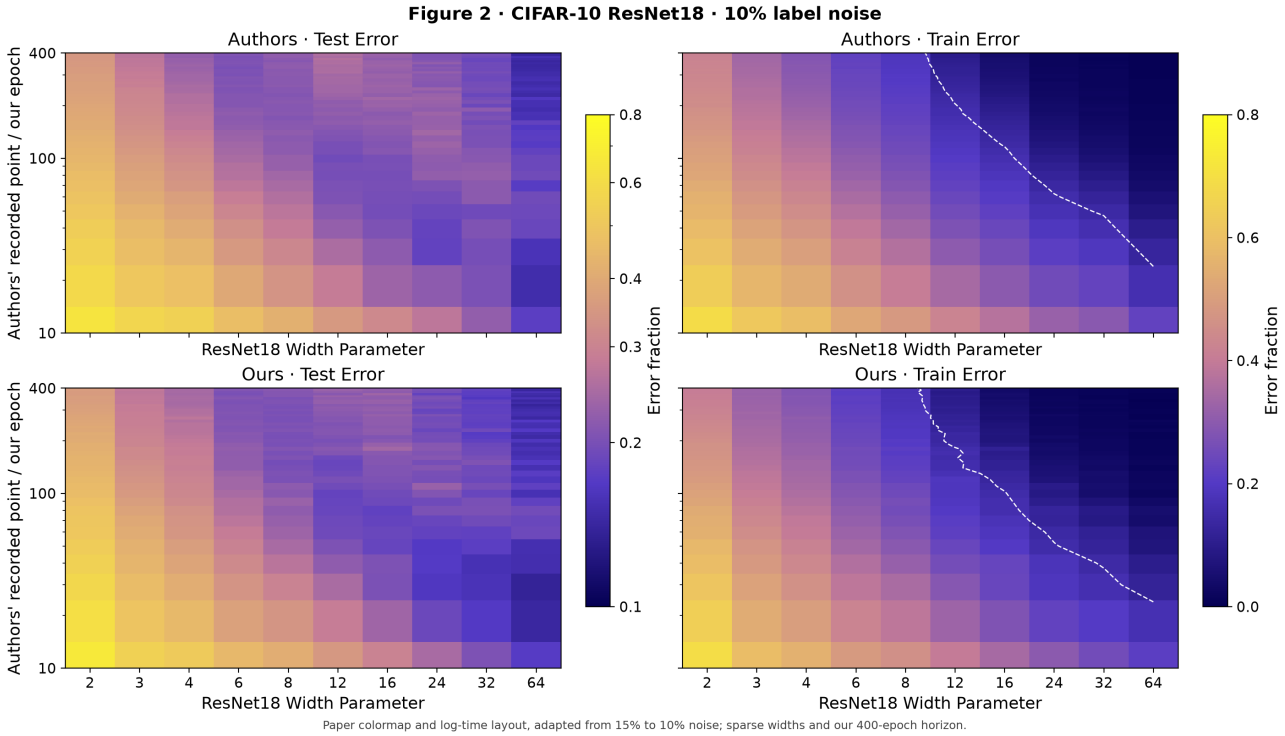
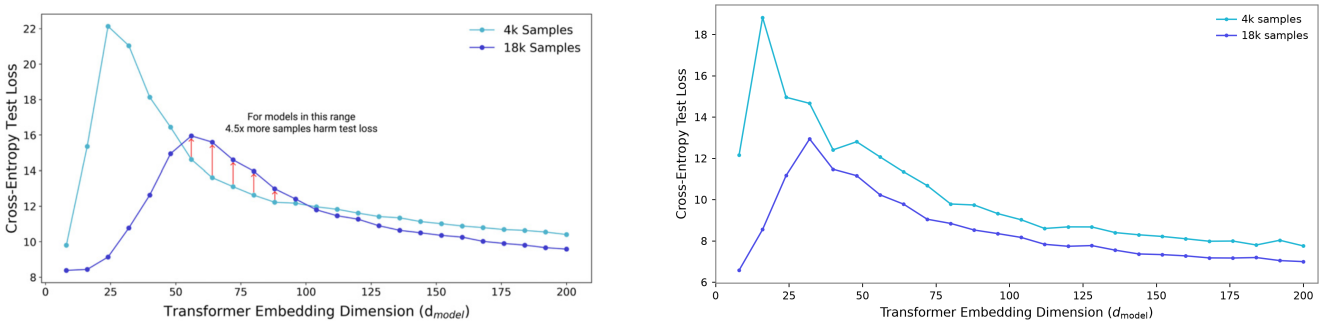


Fig. 4. Train and test heatmaps (paper Figure 2): authors' and our CIFAR-10 ResNet18 train/test heatmaps at 10% noise, on the same ten widths and displayed horizon. Authors' epoch coordinates follow the released notebook convention; ours are recorded.



(a) Nakkiran et al.

(b) Our reproduction

Fig. 5. Comparison of the published Figure 3 and reproduced model-wise translation experiment on IWSLT'14.

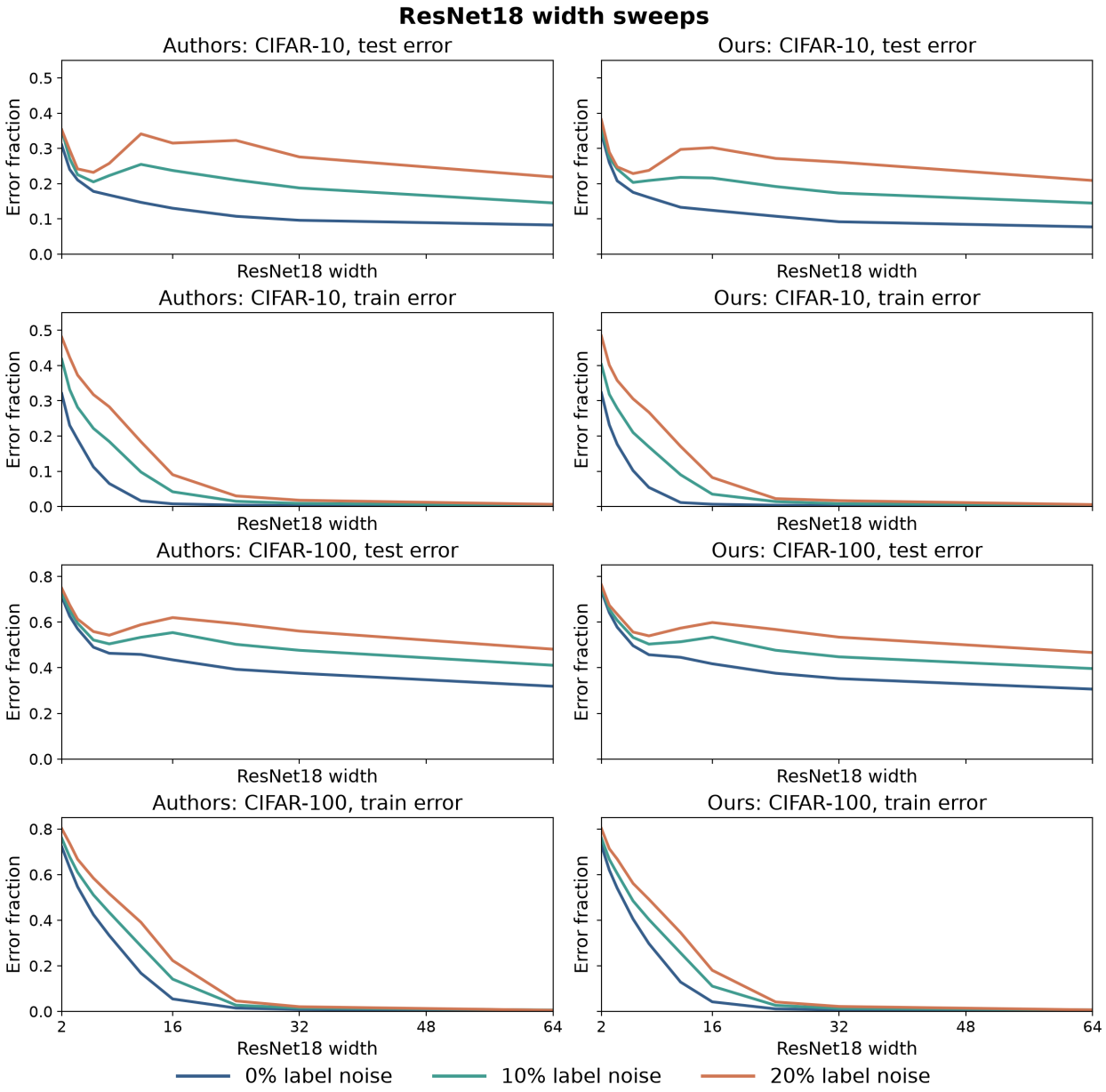


Fig. 6. ResNet18 error versus width on CIFAR-10 and CIFAR-100, reproducing paper Figure 4. Both sources use the same ten widths and label-noise rates of 0, 10, and 20%. The left column shows the authors at displayed epoch 400 (measurement index 399); the right column shows ours at recorded epoch 400.

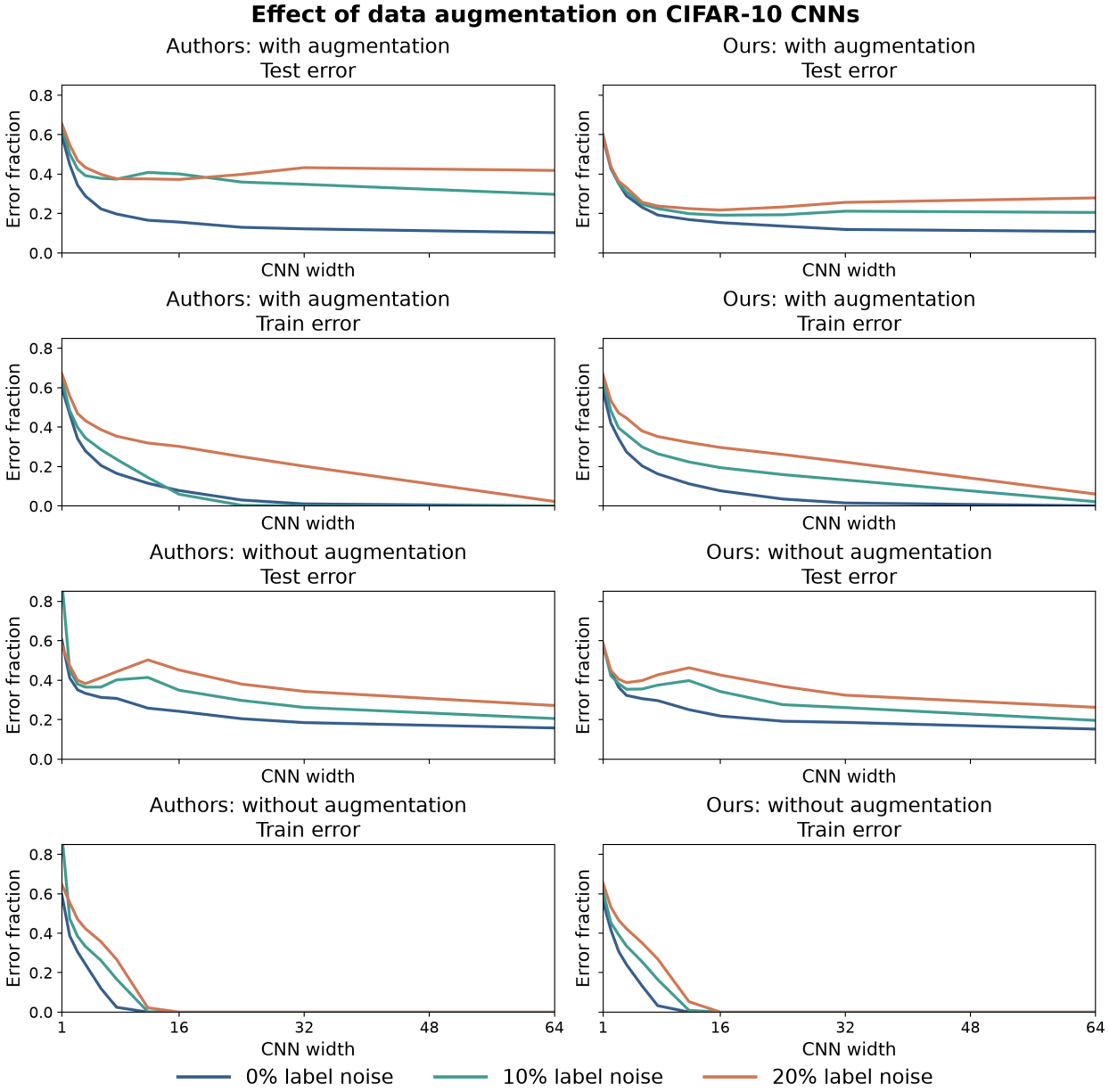


Fig. 7. Effect of data augmentation on CIFAR-10 CNN error, reproducing paper Figure 5. Both sources use the same eleven widths and label-noise rates of 0, 10, and 20%. The left column shows the authors at measurement index 194; the right column shows ours after 50,000 SGD updates. The author point corresponds to approximately 49,920 updates only under the unverified assumption of 256 updates per measurement, so training duration is not confirmed to match.

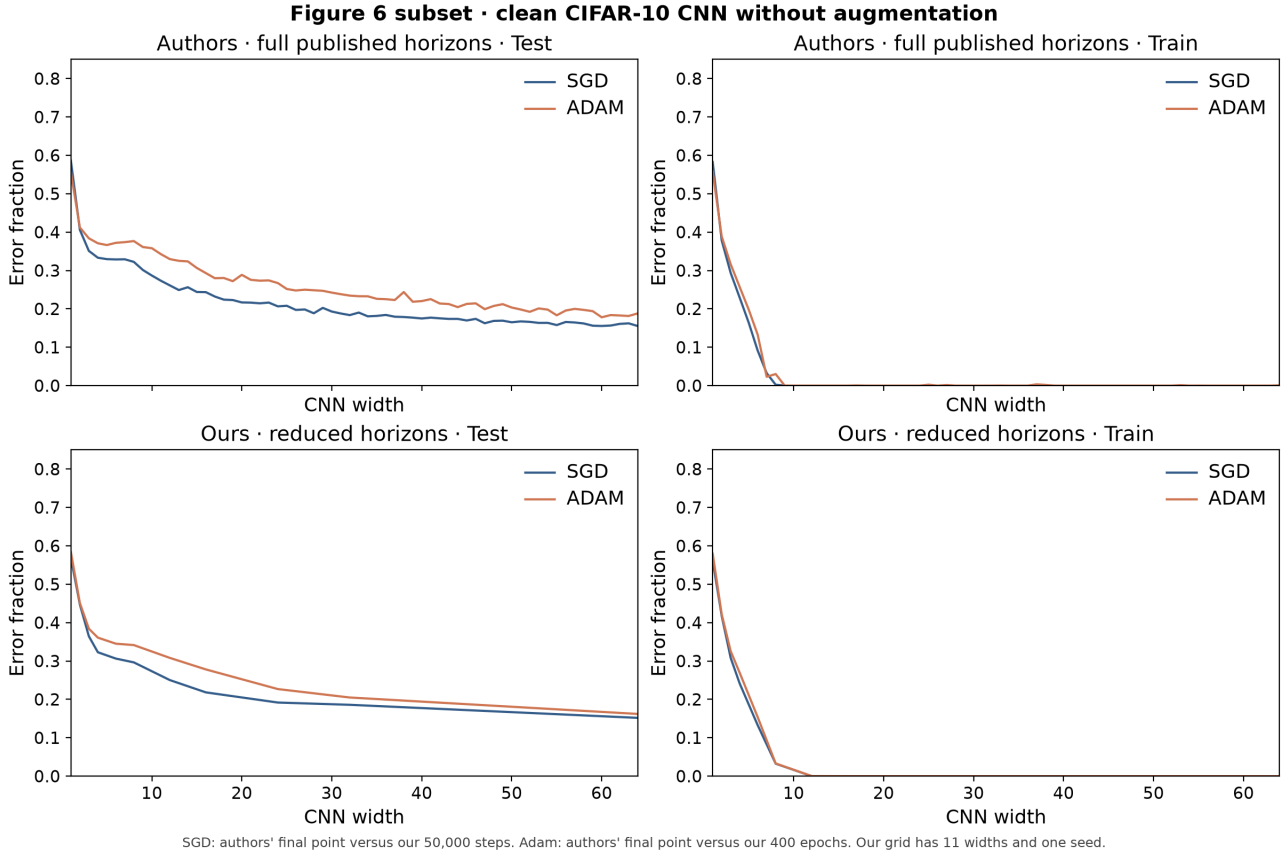


Fig. 8. SGD and Adam comparison (paper Figure 6): authors' and our clean, unaugmented CIFAR-10 CNN results with SGD and Adam. Authors' endpoints use full released histories; ours use 50,000 SGD updates or 400 Adam epochs.

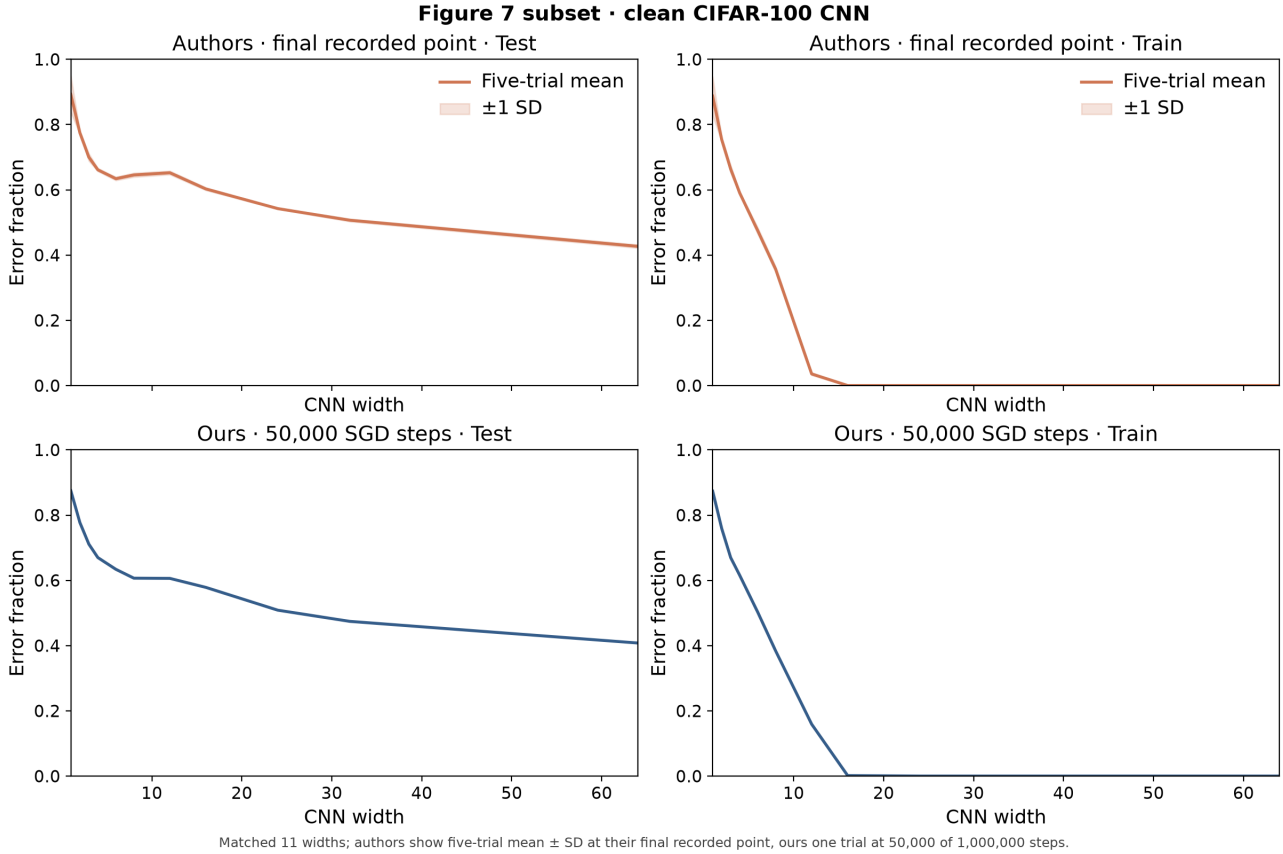


Fig. 9. CIFAR-100 CNN width sweeps (paper Figure 7): authors' and our clean CIFAR-100 CNN results at eleven common widths. The authors' five-trial mean and standard deviation use full histories; ours use one seed and 50,000 SGD updates.

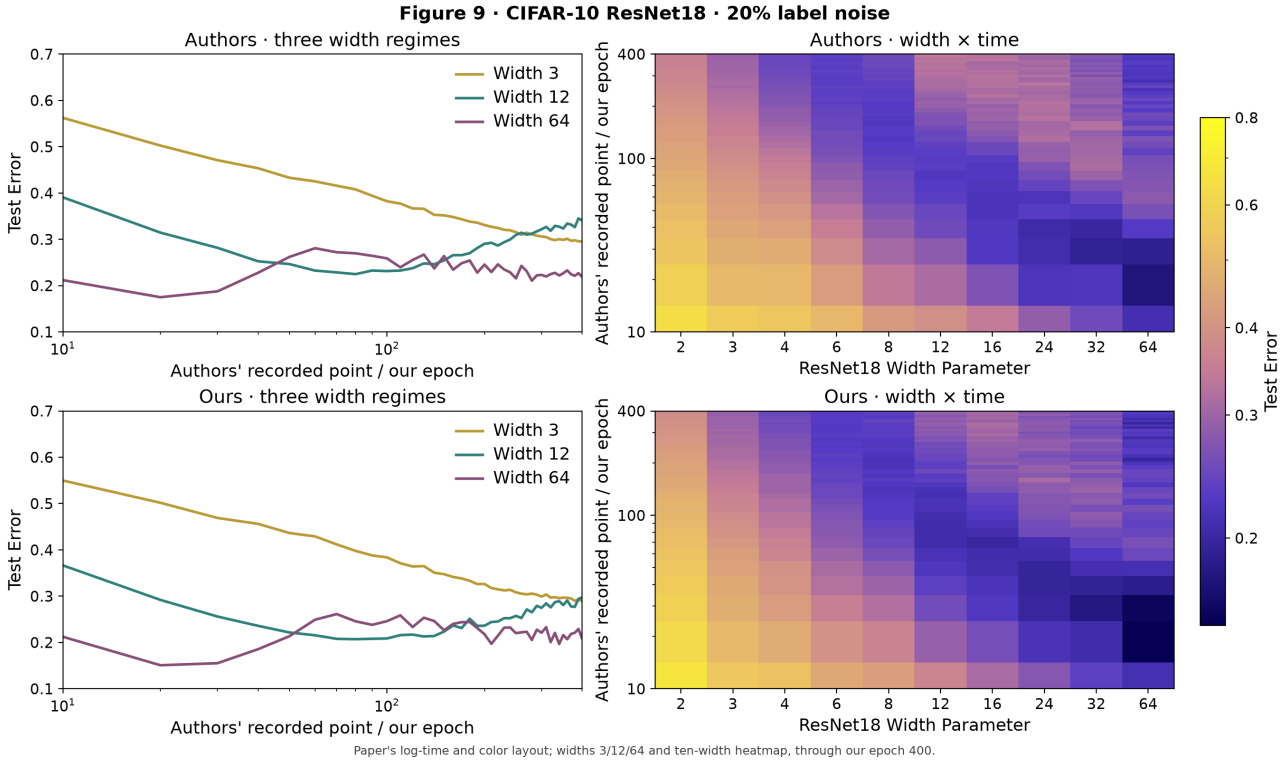


Fig. 10. Epoch-wise CIFAR-10 ResNet18 comparisons (paper Figure 9): three width trajectories and a ten-width heatmap through epoch 400, compared with the authors' released measurements. Our recorded epochs and the authors' notebook epoch convention are distinguished in the plots.

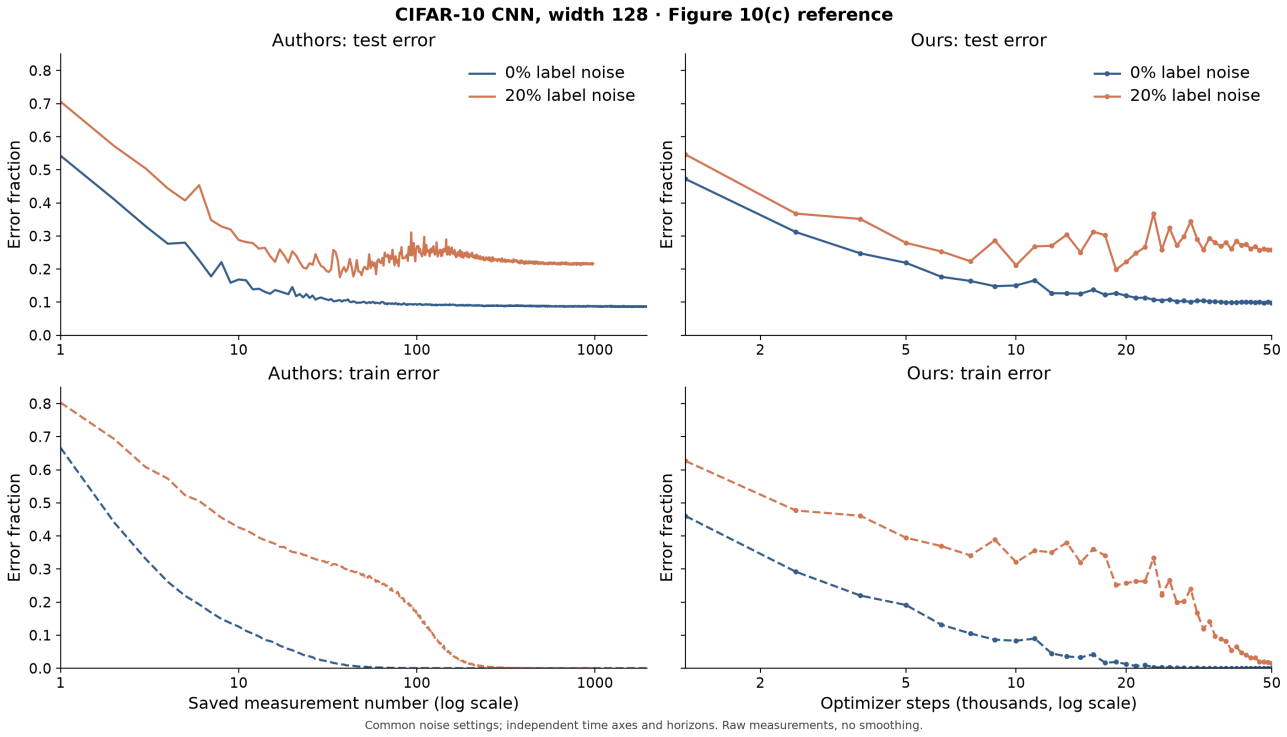


Fig. 11. Training trajectories (paper Figure 10(c)): authors' and our width-128 CIFAR-10 CNN trajectories at the common 0 and 20% noise settings. Authors' clean full-data and 50,000-example/20%-noise references use saved measurement numbers; ours use recorded optimizer steps through 50,000 updates. Logging intervals and horizons are not matched. No smoothing is applied. The unresolved 10%-noise author source is excluded.

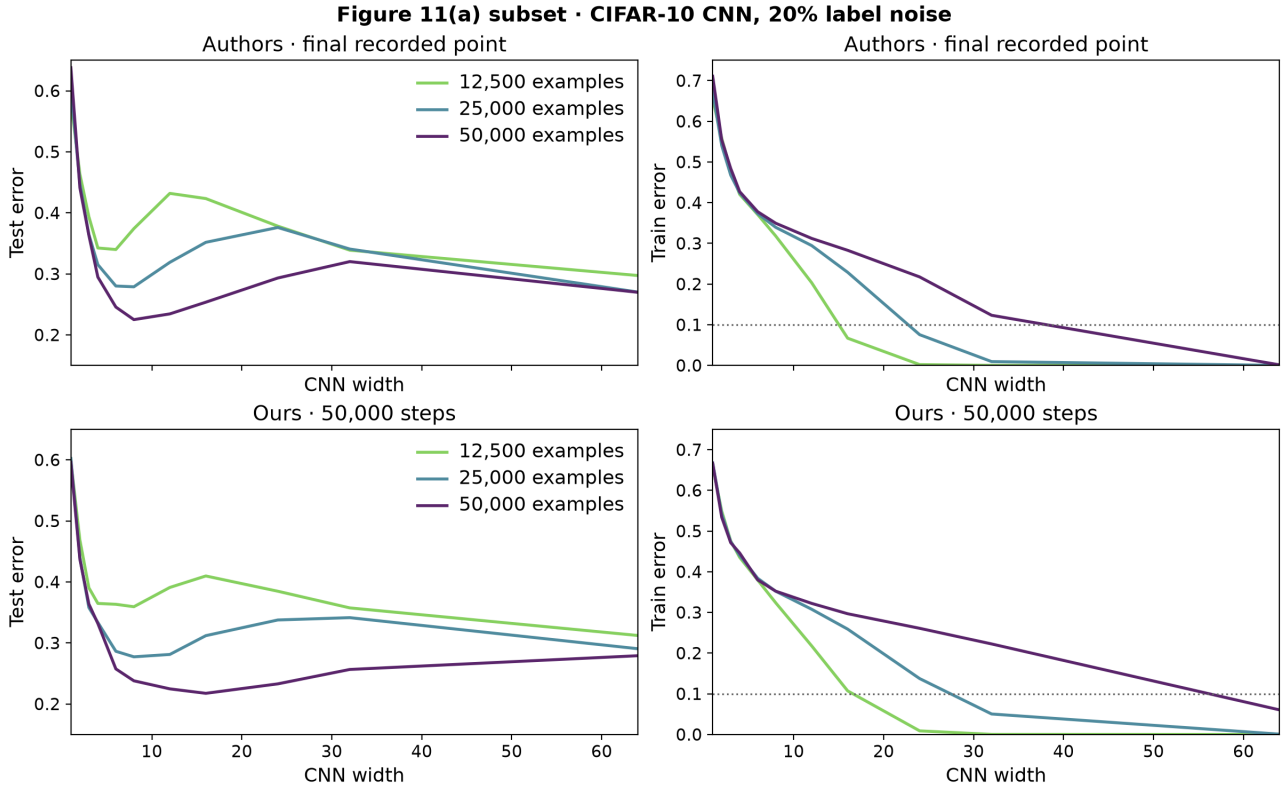
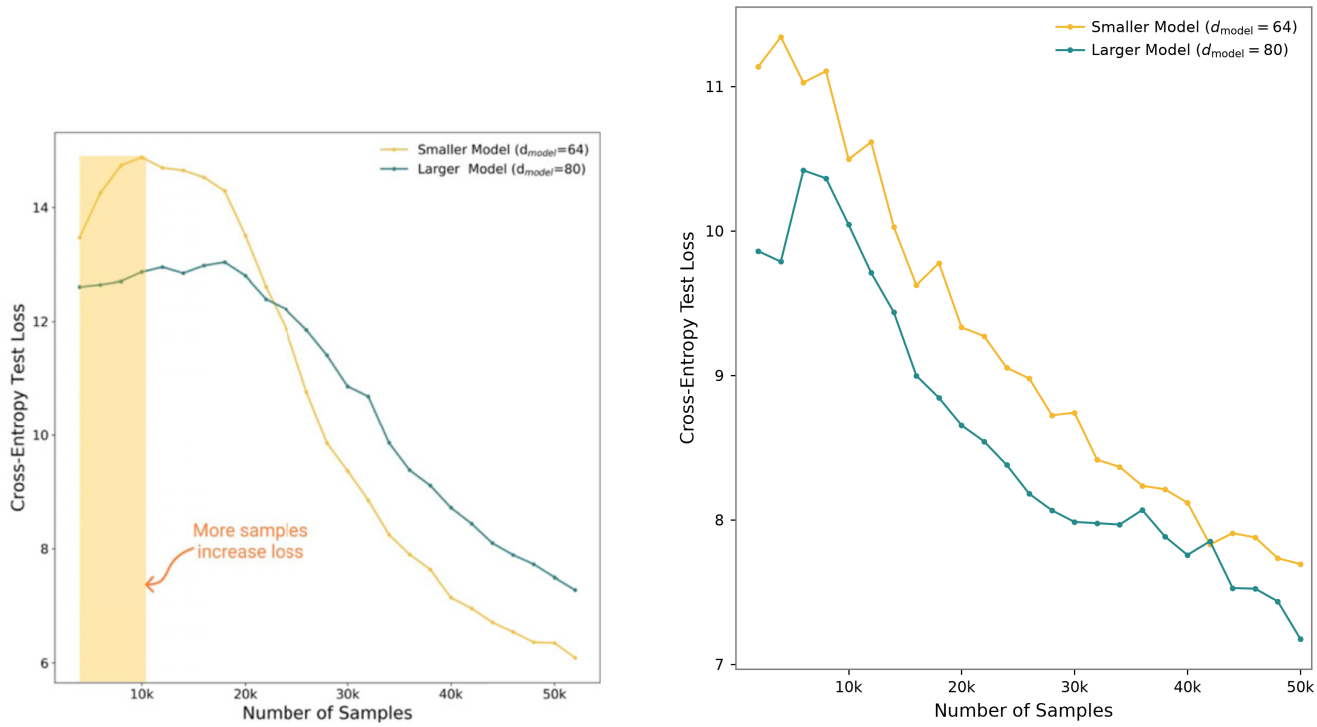


Fig. 12. Effect of training-set size (paper Figure 11(a), 20% panel): authors' and our 20%-noise CIFAR-10 CNN test and train error for 12,500, 25,000 and 50,000 examples at the eleven common widths. Authors' curves come from the released Figure 12 grid, a single trial, whereas the published panel shows the mean and standard deviation of five; ours are one seed at 50,000 updates. Dotted line: train error 0.1. Table 2 uses the finer grid of each side. Our 10%-noise subset runs are excluded because no matching author source was identified.



(a) Nakkiran et al.

(b) Our reproduction

Fig. 13. Comparison of the published Figure 11(b) and reproduced sample-wise translation experiment on IWSLT'14.

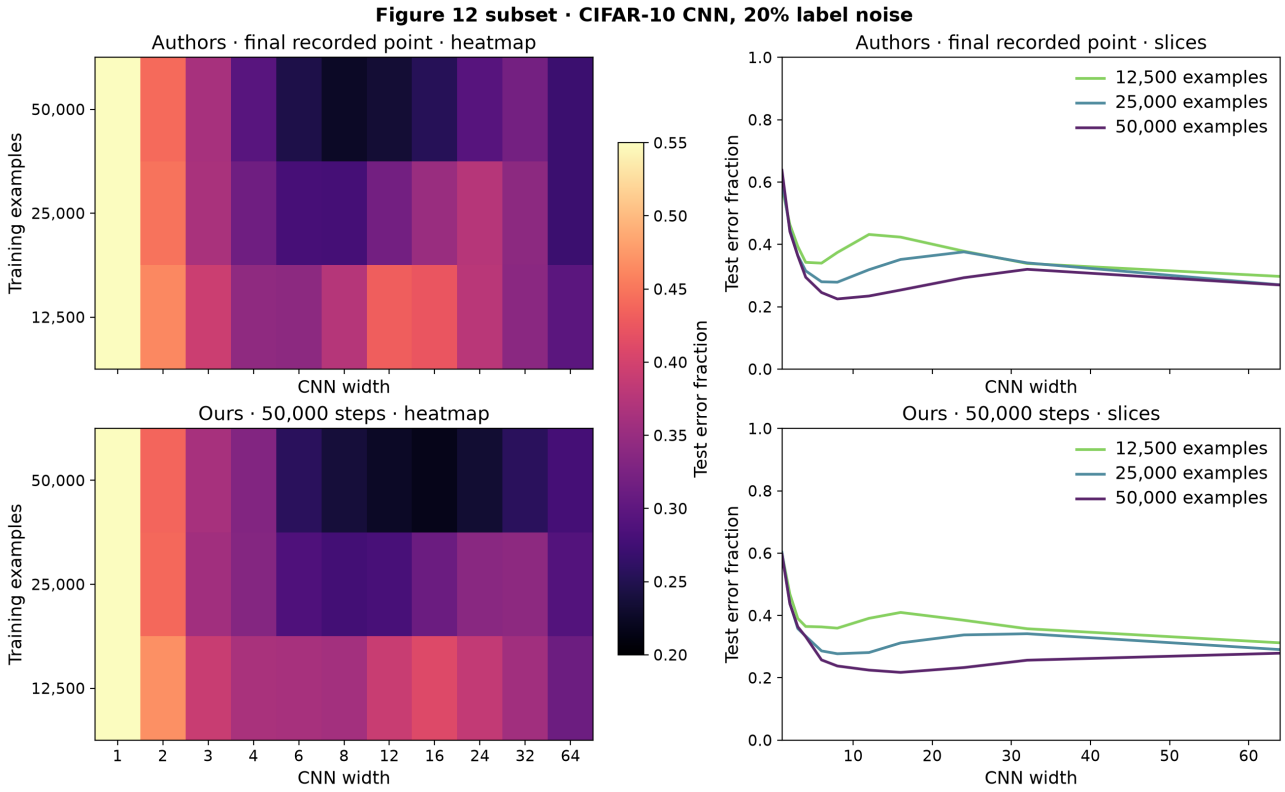


Fig. 14. Sample-size heatmaps and slices (paper Figure 12): authors' and our 20%-noise CIFAR-10 CNN sample-size heatmaps and slices. Both use three sample sizes, eleven common widths, and a common color scale; training horizons differ.